அரசுப் பொதுத் தேர்வு : 2024

பாடம் : கணிதம் – விடை குறிப்பு

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PART – I

TYPE – A

1	a	$\frac{8}{3}$
2	d	$\frac{3\pi a^4}{16}$
3	a	10
4	d	2
5	d	8
6	a	$[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 0$
7	d	$\frac{d^2y}{dx^2} - y = 0$
8	b	$y = 0$
9	d	$\frac{1}{(1+x)^2} dx$
10	a	$x^2 + y^2$
11	c	2
12	d	பரவளையம் / parabola
13	c	$\begin{bmatrix} 5 & -2 \\ 3 & -1 \end{bmatrix}$
14	a	45°
15	d	$\frac{1}{\sqrt{5}}$
16	b	$-\frac{9}{7}$
17	a	0
18	c	2
19	d	$\text{adj}(AB) = (\text{adj}A)(\text{adj}B)$
20	c	-4

PART – II

21

$$\sum_{n=1}^{12} i^n = i^1 + i^2 + \cdots + i^{12}$$

$$= (i^1 + i^2 + i^3 + i^4) + (i^5 + i^6 + i^7 + i^8) + (i^9 + i^{10} + i^{11} + i^{12})$$

$$= 0 + 0 + 0$$

$$= 0$$

22

$\alpha + \beta = \frac{7}{2}$, $\alpha\beta = \frac{13}{2}$

$\alpha^2 + \beta^2 = \frac{-3}{4}$, $\alpha^2\beta^2 = \frac{169}{4}$

$x^2 + \frac{3}{4}x + \frac{169}{4} = 0$

$4x^2 + 3x + 169 = 0$

23

$df = (2x+3)dx$

$x = 3$ எனில் $dx = 0.02$

$df = (6+3)(0.02)$

$= 9(0.02)$

$= 0.18$

24

$y = mx$ —①

$\frac{dy}{dx} = m$

① $\Rightarrow$ $y = x\frac{dy}{dx}$

$\Rightarrow \frac{dy}{dx} = \frac{y}{x}$

25

$E(X) = \int_1^2 f(x)dx = \int_1^2 2(x-1)dx$

$= 2\left[\frac{x^2}{2} - x\right]_1^2$

$E(X) = 1$

சராசரி $= 1$

26

$(h, k) = (-3, -4)$, $r = 3$

$(x-h)^2 + (y-k)^2 = r^2$

$(x+3)^2 + (y+4)^2 = 3^2$

$x^2 + y^2 + 6x + 8y + 16 = 0$

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27 $A = \begin{bmatrix} -1 & 3 \\ 4 & -7 \\ 3 & -4 \end{bmatrix}$, $\rho(A) \le \{\min 3, 2\} = 2$

$\begin{vmatrix} -1 & 3 \\ 4 & -7 \end{vmatrix} = 7 - 12$

$= -5$

$\neq 0$

தரம் $\rho(A) = 2$

28 $\int_0^{\frac{\pi}{2}} \sin^{10} x \, dx = \frac{9}{10} \times \frac{7}{8} \times \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2}$

$= \frac{63\pi}{512}$

29 $\lim_{x \to 1} \frac{x^2 - 3x + 2}{x^2 - 4x + 3}$

$= \lim_{x \to 1} \left(\frac{2x - 3}{2x - 4} \right)$

$= \frac{2-3}{2-4} = \frac{-1}{-2}$

$= \frac{1}{2}$

30 $[\vec{a}\ \vec{b}\ \vec{c}] = \begin{vmatrix} 2 & -1 & 3 \\ 1 & -1 & 0 \\ 3 & -1 & 6 \end{vmatrix}$

$= 2(-6-0) + 1(6-0) + 3(-1+3)$

$= -12 + 6 + 6$

$= 0$

இதே தான் வலப்பக்கம்

PART – III

31 $\cos^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right) = \alpha$ எனக்கொள்.

$\cot \alpha = \frac{1}{\sqrt{x^2-1}}$

$\sec \alpha = \frac{x}{1} = x$

$\alpha = \sec^{-1} x$

$\cos^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right) = \sec^{-1} x$

32 $x^2 + 6x + 4y + 5 = 0$

$(x+3)^2 = -4(y-1)$

$X^2 = -4Y$

நாண் தொடர்பு சமன்பாடு

$2x + y + 1 = 0$

செங் தொடுகோடு சமன்பாடு

$x - 2y - 7 = 0$

33 $LHS = [\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}]$

$= (\vec{a} - \vec{b}) \cdot ((\vec{b} - \vec{c}) \times (\vec{c} - \vec{a}))$

$= [\vec{a}\ \vec{b}\ \vec{c}] - 0 + 0 - 0 + 0$

$- [\vec{b}\ \vec{c}\ \vec{a}]$

$= 0$

$= RHS$

34 $u(x, y) = \frac{x^2 + y^2}{\sqrt{x+y}}$

$u(\lambda x, \lambda y) = \frac{\lambda^2 x^2 + \lambda^2 y^2}{\sqrt{\lambda}\sqrt{x+y}}$

$= \lambda^{2 - \frac{1}{2}} u(x, y)$

$= \lambda^{\frac{3}{2}} u(x, y)$

$u(x, y)$-ன் படி $\frac{3}{2}$ உடைய சமபடித்தானது

சார்பு: $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$

$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{3}{2} u$

35 $x + y = 12$ —— (1)

$y = 12 - x$

$P = xy = x(12 - x) = 12x - x^2$

$\frac{dP}{dx} = 12 - 2x$

$\frac{dP}{dx} = 0 \Rightarrow 12 - 2x = 0$

$x = 6$

(1) $\rightarrow$ $y = 12 - 6$

$y = 6$

பெருமம் எனும் சார் $x = 6$, $y = 6$

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36 $I = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{1}{1+\sqrt{\tan x}}\, dx$ —— ①

$I = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{\sqrt{\tan x}}{1+\sqrt{\tan x}}\, dx$ —— ②

Adding ①, ②

$I + I = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{1+\sqrt{\tan x}}{1+\sqrt{\tan x}}$

$2I = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} 1\, dx$

$= \left[x\right]_{\frac{\pi}{8}}^{\frac{3\pi}{8}}$

$2I = \frac{\pi}{4}$

$I = \frac{\pi}{8}$

37 $\frac{1+i}{1-i} = i, \quad \frac{1-i}{1+i} = -i$

$\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3 = i^3 - (-i)^3$

$= -i - i$

$= -2i$

38 $\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$

$\int \frac{1}{1+y^2} dx = \int \frac{1}{1+x^2} dx$

$\tan^{-1} y = \tan^{-1} x + c$

$\tan^{-1}\left(\frac{y-x}{1+xy}\right) = c$

$\frac{y-x}{1+xy} = \tan c = C$

39 நிகழ்தகவு பரவல் அட்டவணை

x	0	1	2	3
$f(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$f(x) = \begin{cases} \frac{1}{8}, & x = 0, 3 \\ \frac{3}{8}, & x = 1, 2 \end{cases}$

40 $|adj(adj A)| = |A|^{(n-1)^2}$

$\begin{vmatrix} 2 & -1 & 3 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{vmatrix} = -1$

$|adj(adj A)| = (-1)^4 = 1$

PART — IV

41 (a) குறைந்த புள்ளி $\left(\frac{3}{2}, \frac{9}{4}\right)$

$y = x^2$

$\frac{dy}{dx} = 2x, \quad m_1 = 3$

$y = (x-3)^2$

$\frac{dy}{dx} = 2(x-3), \quad m_2 = -3$

$\tan\theta = \left|\frac{3-(-3)}{1-9}\right| = \frac{3}{4}$

$\theta = \tan^{-1}\left(\frac{3}{4}\right)$

b $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$

$\frac{\frac{x-1}{x-2} + \frac{x+1}{x+2}}{1 - \left(\frac{x-1}{x-2}\right)\left(\frac{x+1}{x+2}\right)} = 1$

$2x^2 - 4 = -3$

$x^2 = \frac{1}{2}$

$x = \pm \frac{1}{\sqrt{2}}$

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42 a

தோன்றும் எண்கள்	2	3	4	5	6
இதற்கான எண்ணிக்கை	1	4	10	12	9

நிகழ்தகவு பரவல் அட்டவணை

x	2	3	4	5	6
$f(x)$	$\frac{1}{36}$	$\frac{4}{36}$	$\frac{10}{36}$	$\frac{12}{36}$	$\frac{9}{36}$

இயல்பு பரவல் சார்பு

$$f(x) = \begin{cases} 0 ; & -\infty < x < 2 \\ \frac{1}{36} , & 2 \le x < 3 \\ \frac{5}{36} , & 3 \le x < 4 \\ \frac{15}{36} ; & 4 \le x < 5 \\ \frac{27}{36} ; & 5 \le x < 6 \\ 1 ; & 6 \le x < \infty \end{cases}$$

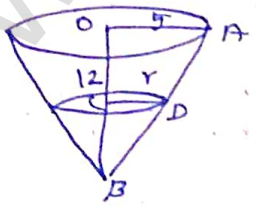
b $\operatorname{Im}\left(\frac{(2x+1)+2iy}{(1-y)+ix} \times \frac{(1-y)-ix}{(1-y)-ix}\right) = 0$

$$\frac{(2x+1)(-x)+2y(1-y)}{(1-y)^2+x^2} = 0$$

$$-2x^2 - x + 2y - 2y^2 = 0$$

$$2x^2 + 2y^2 + x - 2y = 0$$

43 (a) $12r = 5h$



$r = \frac{5h}{12}$

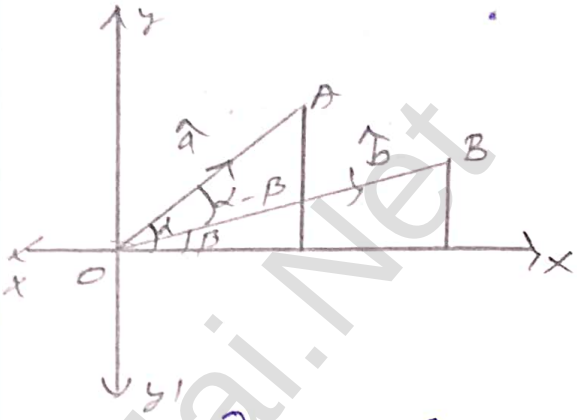
$V = \frac{1}{3}\pi r^2 h$

$V = \frac{25\pi h^3}{\frac{1}{3} \times 144}$

$\frac{dv}{dt} = \frac{25\pi}{144} h^2 \frac{dh}{dt}$

$\frac{dh}{dt} = \frac{10 \times 144}{25\pi \times 64} = \frac{9}{10\pi}$ மீ/நிமிடம்

b $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$



$\hat{a} = \cos\alpha\,\hat{i} + \sin\alpha\,\hat{j}$

$\hat{b} = \cos\beta\,\hat{i} + \sin\beta\,\hat{j}$

$\hat{b} \times \hat{a} = (\sin\alpha\cos\beta - \cos\alpha\sin\beta)\hat{k}$ — (1)

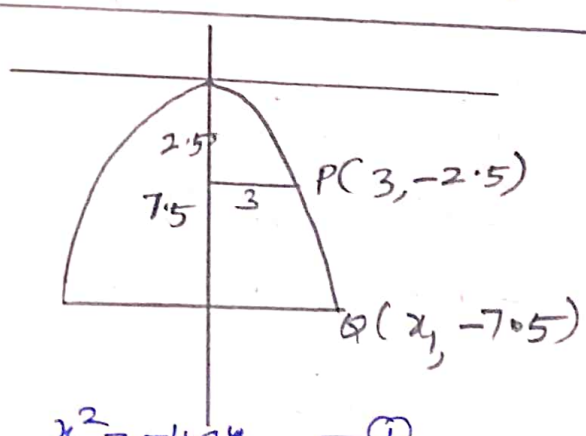
$\hat{b} \times \hat{a} = |\hat{b}||\hat{a}|\sin(\alpha-\beta)\hat{k}$

$= \sin(\alpha-\beta)\hat{k}$ — (2)

(1), (2)

$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

44 (a)



$x^2 = -4ay$ — (1)

$(3, -2.5) \Rightarrow 3^2 = -4a(-2.5)$

$4a = \frac{9}{2.5}$

$x^2 = -\frac{9}{25} y$ — (2)

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$(x_1, -7\sqrt{5})$ ② ⇒

$x_1^2 = 9(3)$

$x_1 = 3\sqrt{3}$ மீ

இந்த இடத்திலிருந்து $3\sqrt{3}$ மீ தூரத்திற்கு அப்பால் நீரானது தரையில் விழும்.

(b) $P = \frac{1}{x}$, $Q = \sin x$

$P = \log x$, $I.F = x$

$y e^{\int P dx} = \int Q e^{\int P dx} dx + c$

$xy = \int \sin x \cdot x \, dx + c$

$xy = -x\cos x + \sin x + c$

$xy + x\cos x = \sin x + c$

45 (a) $\int \frac{dA}{A} = A \int dt$

$A = ce^{kt}$ —— ①

$t=0$ எனில் $A = A_0$

$A = A_0 e^{kt}$ —— ②

$t=5$ எனில் $A = 3A_0$

$e^{5k} = 3$

$t=10$ எனில்

$A = 9A_0$

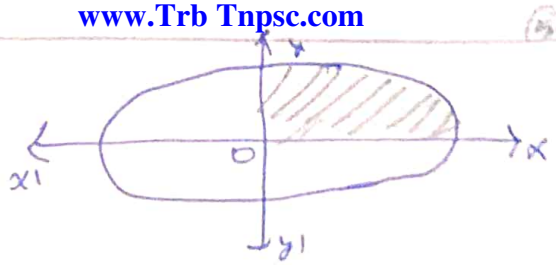
b $\vec{r} = \vec{a} + s\vec{u} + t\vec{v}$

$\vec{r} = (2\hat{i}+2\hat{j}+\hat{k}) + s(2\hat{i}+3\hat{j}+3\hat{k}) + t(3\hat{i}+2\hat{j}+\hat{k})$

C.E $\begin{vmatrix} x-2 & y-2 & z-1 \\ 2 & 3 & 3 \\ 3 & 2 & 1 \end{vmatrix} = 0$

$3x - 7y + 5z + 3 = 0$

46 (a)



$A = 4\int_0^a \frac{b}{a}\sqrt{a^2 - x^2}\, dx$

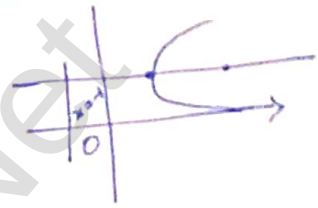
$= \pi ab$

(b) முனை : $(1, 2)$

குவியம் : $(3, 2)$

இயக்குவரை : $x = -1$

செவ்வகலத்தின் நீளம் $= 8$



47 (a)

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

$((\sim P) \vee Q) \wedge ((\sim Q) \vee P) = TFFT$

(b) $\Delta = -15$, $\Delta_1 = -15$,

$\Delta_2 = -5$, $\Delta_3 = -5$

$x = 1$, $y = 3$, $z = 3$

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